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Competition and manipulation in derivative contract markets

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1 Big picture

- **Question.** How should we understand manipulation in derivative contract markets when the settlement price of a contract is determined by trading in an underlying spot market?
- **Core problem.** Derivatives are useful because they let agents hedge exposure to a risk factor, but the derivative payoff is not usually tied directly to the abstract risk factor. It is tied to a price benchmark produced by a spot market. If some derivative traders also trade in that spot market, they may move the benchmark in the direction that benefits their derivative positions.
- **Main definition.** Contract-market manipulation is the component of spot-market trading that is motivated by derivative-contract payoffs. It is distinct from ordinary price impact, liquidity withholding, and bid shading.
- **Main mechanism.** A spot trader with a long contract position wants a high settlement price and therefore buys in the spot auction. A trader with a short contract position wants a low settlement price and therefore sells in the spot auction. Because these trades are driven by derivative payoffs rather than spot fundamentals, they add non-fundamental noise to the settlement price.
- **Main friction.** Agents cannot contract directly on the risk factor ψ ; they must contract on the spot auction price p . In a competitive spot market, $p = \psi$ and derivatives perfectly share risk. In a strategic spot market, p differs from ψ because spot traders strategically bid to affect derivative payoffs.
- **Main equilibrium object.** Manipulation-induced basis risk is

$$\text{Var}(p - \psi),$$

the non-fundamental variance in the settlement price relative to the true risk factor.

- **Market-structure result.** Basis risk falls when total spot-market storage capacity rises, when the spot market becomes more competitive, and when spot traders' derivative-contract positions become smaller.
- **Welfare result.** Manipulation can be Pareto-disimproving: even spot traders, who manipulate and receive transfers on average, can be worse off because other traders' manipulation makes the settlement price risky.
- **Policy implication.** Position limits on derivative holdings of spot-market participants can reduce manipulation risk. The paper gives regulators a way to define manipulation and a way to measure manipulation-induced basis risk using information about bid slopes, contract positions, and inventory shocks.
- **Why the paper matters.** Legal definitions often emphasize intent to move prices, but intentional price impact is present in almost every strategic market. The paper offers a sharper economic definition: manipulation is the part of spot trading that would disappear if the trader did not hold derivative contracts tied to the benchmark.

2 Institutional background

2.1 Derivative contracts and settlement benchmarks

Derivative contracts transfer exposure to a risk factor. Examples include equity-index futures, interest-rate swaps, volatility futures, energy futures, commodity futures, and many cash-settled contracts whose payoff is tied to a benchmark price. The benchmark is usually constructed from prices in a spot market or from a price-indexing mechanism connected to spot trades.

- A cash-settled derivative pays according to a benchmark price rather than physical delivery of the underlying good.
- If the benchmark accurately reflects the risk factor, the contract can be used to hedge the risk factor.
- The problem arises because the benchmark is endogenous to the behavior of traders in the spot market.
- If a trader simultaneously holds a derivative position and trades in the spot market, spot-market trading can affect the derivative payoff.

Interpretation: The settlement benchmark is supposed to be a measurement device, but it also becomes a strategic target. This is the core reason manipulation can exist even when all trades are real trades rather than fictitious quotes.

2.2 Liquidity mismatch between contract and spot markets

The paper stresses that derivative markets can be much larger than the spot markets used to settle them. Contract markets often provide cheap and liquid exposure to risks, while spot markets are costly because they involve physical storage, balance-sheet capacity, capital requirements, logistics, or other frictions.

- Agents prefer derivatives because derivatives avoid direct spot-market holding costs.
- Spot markets may have limited storage or balance-sheet capacity.
- A large derivative position can create incentives to trade a relatively small spot market aggressively near settlement.
- This mismatch makes settlement prices vulnerable to strategic pressure.

Interpretation: The manipulation problem is strongest when the contract market is deep but the spot market is thin. Then small spot trades can move a benchmark that determines large derivative payoffs.

2.3 Legal background and the definitional problem

Regulators often define illegal manipulation as trading with the intent to create artificial prices. The paper argues that this definition is too broad because almost all strategic traders understand that their trades move prices. Intentional price impact alone does not distinguish illegal manipulation from ordinary strategic trading.

- A trader may legally shade bids or manage price impact when trading a large order.
- A manipulator trades the spot good to improve payoffs on a separate derivative contract.
- Legal cases often rely on communications showing intent, but such evidence is incomplete and difficult to interpret.
- An economic definition should separate manipulation from standard market power or optimal execution.

Interpretation: The paper's contribution is partly conceptual. It does not say every price-impacting trade is manipulation. It identifies a specific cross-market externality: spot trades distort a settlement price that determines derivative payoffs.

3 Model setup

3.1 Agents and timing

There are three kinds of agents:

- a representative pure hedger H , who trades derivatives but cannot trade the spot good;
- $n > 2$ spot traders, indexed by $i = 1, \dots, n$, who trade both derivatives and the spot good;
- a risk-neutral competitive market maker who trades derivatives but cannot trade the spot good.

Agents play a five-stage game:

1. the hedger and spot traders draw factor exposures x_H and x_i ;
2. agents submit derivative orders $c_H(x_H)$ and $c_i(x_i)$;
3. the risk factor ψ is realized;
4. spot traders submit bid curves in a uniform-price double auction;
5. derivative contracts settle using the spot auction price p .

Combining all terms, a spot trader's total wealth, considering her factor exposures and outcomes in the contract and spot markets, is:

$$W_{spot}(x_i, c_i, z_i, p, \psi) = \underbrace{\psi x_i - \mu_\psi x_i}_{\text{Factor exposure}} - \underbrace{p_c c_i}_{\text{Contract price}} + \underbrace{p c_i}_{\text{Contract payoff}} + \underbrace{z_i \psi - \frac{1}{2\kappa} z_i^2}_{\text{Spot good payoff}} - \underbrace{p z_i}_{\text{Spot good price}} \quad (3)$$

The pure hedger's total wealth is:

$$W_{hedger}(x_H, c_H, p, \psi) = \underbrace{\psi x_H - \mu_\psi x_H}_{\text{Factor exposure}} - \underbrace{p_c c_H}_{\text{Contract price}} + \underbrace{p c_H}_{\text{Contract payoff}} \quad (4)$$

3.1. Benchmarks

To illustrate how the two frictions, noncontractibility and imperfect competition, prevent agents from perfectly sharing risk, we analyze two benchmark cases in which the first-best outcome is attainable: when agents can directly contract on ψ , and when contracts settle to spot market prices, but spot market participants bid competitively.

Contracting directly on ψ . Agents could perfectly share risk if they could purchase contracts tied directly to the risk factor – that is, contracts which pay ψ per contract in stage 5. In the first period, the market maker would set the price of these contracts equal to their expected payoff:

$$p_c = E[\psi] = \mu_\psi$$

The market maker is willing to buy or sell an infinite amount of contracts at this price, since the hedger and spot traders' endowments and contract purchases are uninformative about ψ . An agent with contract position c_i would receive total wealth:

$$\underbrace{x_i \psi - x_i \mu_\psi}_{\text{Factor exposure}} + \underbrace{\psi c_i}_{\text{Contract payoff}} - \underbrace{p_c c_i}_{\text{Contract price}}$$

Agents play a 5-stage game.

1. The hedger and spot traders draw their factor risk exposures, respectively, x_H and $x_1 \dots x_n$.
2. The hedger and spot traders submit market orders, respectively $c_H(x_H)$ and $c_i(x_i)$, to trade derivative contracts with the market maker.
3. The risk factor, ψ , is realized.
4. Spot traders bid to trade the spot good in an auction.
5. Derivative contracts pay all contract holders based on the spot auction price.

In stage 1, the hedger is endowed with x_H units of a productive asset, and each spot trader i is endowed with x_i units of the productive asset, where:

$$x_H \sim N(0, \sigma_H^2), \quad x_i \sim N(0, \sigma_x^2)$$

A position x_i in the productive asset pays its holder $x_i \psi$ net units of wealth. Thus, the value of each unit of the productive asset depends on a normally distributed risk factor ψ .

(a) Timing and factor exposures

(b) Wealth functions and benchmark cases

Original paper excerpt:

Interpretation: The timing makes the manipulation incentive sharp. Derivative positions are chosen before the risk factor and spot auction. Once ψ is realized, spot traders know both the benchmark-relevant fundamental and their own contract positions, so they can trade the spot good to move settlement prices.

3.2 Preferences and factor exposures

All hedgers and spot traders have CARA utility:

$$U_i(W) = -e^{-\alpha W}.$$

The risk factor is normally distributed:

$$\psi \sim N(\mu_\psi, \sigma_\psi^2).$$

The hedger has exposure $x_H \sim N(0, \sigma_H^2)$, and spot trader i has exposure $x_i \sim N(0, \sigma_x^2)$. A position x_i in the productive asset pays $x_i \psi$. The model normalizes initial wealth so that expected wealth is zero regardless of factor exposure.

Interpretation: The CARA-normal structure lets the paper solve for linear demand and bidding strategies. Economically, the normal risk factor represents any fundamental variable that agents want to hedge, such as energy prices, interest rates, index levels, or volatility.

3.3 Derivative contracts

Agent i chooses a derivative position c_i . In the baseline model, contracts are cash-settled: the holder receives $c_i p$, where p is the spot-auction settlement price. The market maker sets the contract price p_c . In equilibrium, because spot traders are individually small relative to the hedger in the derivative market, the market maker sets

$$p_c = \mu_\psi.$$

Interpretation: The derivative market itself is competitive and frictionless. The distortion comes from the settlement price, not from the derivative-market price.

3.4 Spot-market payoff and storage capacity

If spot trader i buys net quantity z_i of the spot good, the spot payoff is

$$\psi z_i - \frac{1}{2\kappa} z_i^2.$$

The parameter κ measures spot-market storage or holding capacity. A larger κ means lower marginal holding cost and a more elastic spot market.

Interpretation: Storage capacity is the key primitive linking spot-market structure to manipulation risk. When κ is small, even moderate manipulative orders can strongly affect the settlement price.

3.5 Competitive benchmarks

There are two useful benchmarks.

Benchmark 1: Direct contracting on ψ . If agents could contract directly on ψ , the market maker would set $p_c = \mu_\psi$, and each agent would perfectly hedge by choosing $c_i = -x_i$.

Benchmark 2: Competitive spot bidding. If agents cannot contract on ψ but spot traders bid competitively, each spot trader submits

$$z_{Bi}(p) = \kappa(\psi - p).$$

The spot auction clears at $p = \psi$, so the contract again perfectly hedges the risk factor.

Interpretation: The first benchmark removes the noncontractibility friction. The second leaves the noncontractibility friction but removes strategic spot-market behavior. Manipulation requires both frictions: imperfect benchmarking and strategic spot-market trading.

4 Equilibrium of the baseline model

4.1 Proposition 1: equilibrium bid curves and contract positions

Proposition 1 gives the unique symmetric linear equilibrium. Spot traders submit bid curves

$$z_{Bi}(p; c_i, \psi) = \frac{1}{n-1} c_i - \frac{n-2}{n-1} \kappa(p - \psi).$$

The spot auction price satisfies

$$p - \psi = \frac{\sum_{i=1}^n c_i}{n(n-2)\kappa} \sim N\left(0, \frac{\sigma_c^2}{n(n-2)^2\kappa^2}\right).$$

Spot traders' contract positions are linear in their factor exposures:

$$c_i(x_i) = -tx_i, \quad \sigma_c^2 = t^2 \sigma_x^2.$$

sitions; thus, colluding spot traders, who coordinate their spot market bids to maximize joint profits, would manipulate more per unit contract than independently optimizing traders. This is an interesting direction to explore, but I leave this to future work.

4. Equilibrium

Proposition 1 describes equilibrium values of spot traders' bid curves, auction prices, contract purchases, and expected utility for all agents.

Proposition 1. For any $\alpha, \mu_\psi, \sigma_\psi^2, \kappa, \sigma_x^2, \sigma_H^2, n$, there is a unique equilibrium, in which spot traders' spot market bids and contract purchasing strategies are linear and symmetric across agents. In stage 1, the market maker sets contract price $p_c = \mu_\psi$. Spot traders' bid curves in the spot market are:

$$z_{Bi}(p; c_i, \psi) = \frac{1}{n-1} c_i - \frac{n-2}{n-1} \kappa (p - \psi) \quad (7)$$

Spot auction prices are:

$$p - \psi = \frac{\sum_{i=1}^n c_i}{n(n-2)\kappa} \sim N\left(0, \frac{\sigma_c^2}{n(n-2)^2 \kappa^2}\right) \quad (8)$$

Spot traders' contract positions $c_i(x_i)$ are linear in traders' factor exposures x_i , so spot traders' contract positions are normally distributed with mean 0 and variance σ_c^2 , where:

$$c_i(x_i) = -tx_i, \quad \sigma_c^2 = t^2 \sigma_x^2 \quad (9)$$

where t satisfies:

$$t \equiv \left(1 + \frac{\alpha \sigma_\eta^2 - \kappa}{(\alpha \sigma_\psi^2 \kappa)((n^2 - 2n)\kappa + \alpha \sigma_\eta^2)}\right)^{-1} \quad (10)$$

and σ_η^2 is the unique positive value satisfying:

(a) Proposition 1: baseline strategic equilibrium

$$\sigma_\eta^2 > \frac{\kappa - (\alpha \sigma_\psi^2)(n^2 - 2n)\kappa^2}{\alpha(1 + \alpha \sigma_\psi^2 \kappa)} \quad (12)$$

Spot traders' expected utility, as a function of t , is:

$$-\sqrt{\frac{(n^2 - 2n)\kappa}{\alpha\left(\frac{t^2 \sigma_x^2}{n-1}\right) + (n^2 - 2n)\kappa}} \sqrt{1 - \alpha \sigma_x^2 \left(\alpha \sigma_\psi^2 (1-t)^2 + \left(\frac{\alpha\left(\frac{t^2 \sigma_x^2}{n-1}\right) - \kappa}{(\alpha \sigma_\psi^2 \kappa)((n^2 - 2n)\kappa + \alpha\left(\frac{t^2 \sigma_x^2}{n-1}\right))} \right) t^2 \right)} \quad (13)$$

The hedger's net contract position is linear in the hedger's factor exposure x_H :

$$c_H(x_H) = -\frac{\sigma_\psi^2}{\sigma_\psi^2 + \text{Var}(p - \psi)} x_H = -\frac{\sigma_\psi^2}{\sigma_\psi^2 + \frac{\sigma_c^2}{n(n-2)^2 \kappa^2}} x_H \quad (14)$$

The hedger's wealth has mean 0, and its variance, over uncertainty in ψ , x_H , and all x_i 's, is:

(b) Equations for spot-trader utility and hedger exposure

Original paper excerpt:

Interpretation: The bid curve contains two components. The $-\frac{n-2}{n-1}\kappa(p - \psi)$ term is the ordinary strategic supply/demand component: traders provide less liquidity than in the competitive benchmark. The $\frac{1}{n-1}c_i$ term is the manipulation component: a long contract position increases bids, and a short contract position decreases bids.

4.2 Three distinct forms of strategic behavior

The equilibrium separates three behaviors:

- **Liquidity reduction:** spot traders lower bid-curve slopes relative to the competitive benchmark because they have price impact.
- **Bid shading:** when inventory shocks are introduced later, traders pass inventory shocks through less than one-for-one because they internalize price impact.
- **Contract-market manipulation:** the bid curve depends on c_i , even though c_i does not affect the trader's marginal value of the spot good.

Interpretation: The third component is the paper’s definition of manipulation. It is the part of spot trading that exists because the trader holds a derivative contract linked to the settlement benchmark.

4.3 Manipulation-induced basis risk

The central distortion is

$$\text{Var}(p - \psi) = \frac{\sigma_c^2}{n(n-2)^2\kappa^2}.$$

This variance is called manipulation-induced basis risk.

Original paper excerpt:

Since auction prices depend on spot traders’ contract positions c_i , which depend on traders’ random factor exposures x_i , the equilibrium auction price p is a noisy signal of the risk factor ψ . The difference, $p - \psi$, is normally distributed, and expression (8) characterizes its variance:

$$\text{Var}(p - \psi) = \frac{\sigma_c^2}{n(n-2)^2\kappa^2} \quad (16)$$

I call this variance *manipulation-induced basis risk*.

Comparative statics of basis risk. We will define three quantities that influence the level of basis risk. First, define the aggregate *storage capacity* of the market as $n\kappa$, the sum over all n spot traders’ storage capacities. $n\kappa$ would be the slope of aggregate market demand for the spot good, if all agents submitted bid curves as if they had no price impact, as in (6). Second, define the level of market *competition* simply as n , the number of spot traders present in the market. Third, define the size of contract positions as σ_c^2 , the variance of an individual spot trader’s contract position. This is an equilibrium object in the model, but in practice it can often be observed directly by regulators, and directly influenced using policies such as contract position limits.

Basis risk decreases to 0 as storage capacity becomes infinitely large, $n\kappa \rightarrow \infty$, holding fixed n and σ_c^2 . Intuitively, when $n\kappa$ is large, the price impact of manipulative spot trading decreases towards 0, causing prices converge to ψ . However, while *prices* become competitive, spot market *quantities* do not become competitive. From (7), the coefficient on contract positions in agents’ bids is $\frac{1}{n-1}$, independent of κ . If κ increases holding n fixed, prices converge to ψ , but spot market bids, and thus allocations in the spot market, are still distorted in the limit.

Basis risk also decreases towards 0 as markets become competitive, $n \rightarrow \infty$, holding fixed $n\kappa$ and σ_c^2 . This is be-

Read:

- Basis risk is zero in the competitive benchmark because $p = \psi$.
- Basis risk is positive in the strategic equilibrium because spot traders’ contract positions enter spot-market bids.
- Basis risk falls as κ rises because spot-market price impact declines.
- Basis risk falls as n rises because contract positions affect bids less in more competitive spot markets.
- Basis risk rises with σ_c^2 , the dispersion of spot traders’ derivative positions.

Economic message: Manipulation is not measured by the level of prices alone. It is measured by the extra settlement-price noise generated by derivative-position-driven spot trades.

4.4 Contract-position choice and hedging aggressiveness

Spot traders choose $c_i = -tx_i$. The parameter t measures hedging aggressiveness.

- If $t = 1$, spot traders fully hedge their factor exposure.
- If $0 < t < 1$, spot traders under-hedge because manipulation creates basis risk.
- If $t > 1$, spot traders over-hedge because they anticipate profiting from moving the spot settlement price in favor of their derivative positions.

Interpretation: Manipulation creates two opposing incentives. Basis risk discourages contract use, but expected manipulation rents encourage larger contract positions. This is why equilibrium contract positions can be too small or too large relative to pure hedging needs.

4.5 Spot trader welfare

Spot traders can benefit from moving settlement prices in their favor, but they also suffer because other spot traders' manipulation makes settlement prices noisy. The latter force can dominate.

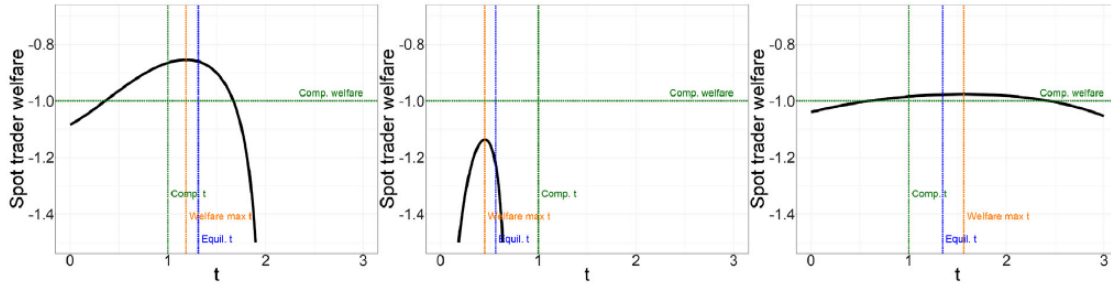


Fig. 1. Spot trader welfare.

Notes. Expected welfare of spot traders, (13), as a function of the hedging aggressiveness parameter t , for three different sets of parameters. The green vertical and horizontal lines denote the competitive values of t and welfare, which are always equal to 1 and -1 respectively. The blue line denotes the unique equilibrium value of t , and the orange line denotes the value of t which maximizes spot traders' welfare. The parameters for the left panel are $n = 3, \alpha = 1, \kappa = 0.5, \sigma_\psi^2 = 0.3, \sigma_x^2 = 0.5$; for the middle panel, they are $n = 5, \alpha = 1, \kappa = 0.05, \sigma_\psi^2 = 0.9, \sigma_x^2 = 1$; for the right panel, they are $n = 3, \alpha = 1, \kappa = 0.8, \sigma_\psi^2 = 1.5, \sigma_x^2 = 0.05$.

Figure 3: Figure 1: spot trader welfare as a function of hedging aggressiveness t

Original paper excerpt:

Read:

- The horizontal green line is competitive welfare.
- The blue vertical line is equilibrium t .
- The orange vertical line is the welfare-maximizing t .
- Across parameterizations, equilibrium can lie to the right of the welfare-maximizing position.
- A reduction in hedging aggressiveness can increase spot-trader welfare.

Economic message: Even the agents who manipulate can be worse off. Each trader wants to exploit her own contract position, but aggregate manipulation creates settlement-price risk for everyone.

4.6 Hedger welfare

The hedger cannot trade the spot good. The hedger's contract position per unit of risk exposure is

$$-\frac{\sigma_\psi^2}{\sigma_\psi^2 + \text{Var}(p - \psi)}.$$

The hedger's wealth variance is

$$\frac{\text{Var}(p - \psi)}{\text{Var}(p - \psi) + \sigma_\psi^2} \sigma_\psi^2 \sigma_H^2.$$

Interpretation: The hedger buys fewer contracts because the settlement price is a noisy signal of the risk factor. The hedger's welfare loss is monotone in manipulation-induced basis risk.

5 Extensions

5.1 Asymmetric spot traders

The baseline assumes symmetric spot traders. The first extension allows trader-specific storage capacity κ_i and arbitrary contract positions. The spot trader's reduced-form wealth is

$$W_{spot,i}(c_i, z_i, p) = pc_i + \psi z_i - \frac{z_i^2}{2\kappa_i} - pz_i.$$

Proposition 2 shows that equilibrium bid curves become

$$z_{Bi}(p; c_i) = \frac{b_i}{\sum_{j \neq i} b_j} c_i - b_i(p - \psi),$$

with the price depending on capacity-weighted contract positions.

pendix B.4 discusses, spot traders may buy contracts to speculate rather than hedge, or as online appendix B.5 discusses, spot traders' contract purchases may have price impact. A spot trader who holds a long contract position profits from higher spot prices, regardless of why she entered into the contract in the first place. Thus, spot traders' contract positions are an observable sufficient statistic for spot traders' incentives to manipulate spot markets.

The following proposition characterizes spot good bids and prices in the unique equilibrium of the general model.

Proposition 2. When spot traders' wealth is described by (19), there is a unique linear ex-post equilibrium, in which i submits the bid curve:

$$z_{Bi}(p; c_i) = \frac{b_i}{\sum_{j \neq i} b_j} c_i - b_i(p - \psi) \quad (20)$$

The spot auction price is:

$$p - \psi = \frac{1}{\sum_{i=1}^n b_i} \sum_{i=1}^n \left[\frac{b_i}{\sum_{j \neq i} b_j} c_i \right] \quad (21)$$

bid slopes b_i satisfy:

$$b_i = \frac{B + 2\kappa_i - \sqrt{B^2 + 4\kappa_i^2}}{2} \quad (22)$$

and $B \equiv \sum_{i=1}^n b_i$ is the unique positive solution to the equation:

$$B = \sum_{i=1}^n \frac{2\kappa_i + B - \sqrt{B^2 + 4\kappa_i^2}}{2} \quad (23)$$

Manipulation incentives depend most directly on the coefficient on c_i , $\frac{b_i}{\sum_{j \neq i} b_j}$, in expression (20). From (22), b_i is increasing in κ_i , so agents with larger storage capacities

(a) Proposition 2: asymmetric storage capacity

Since κ_i represents the spot good storage capacity of agent i , s_i can be thought of as i 's share of total storage capacity in the spot market. Define $s_{\max}$ as the largest agent's capacity share:

$$s_{\max} \equiv \max_i s_i$$

The following proposition derives bounds on the c_i coefficients, $\frac{b_i}{\sum_{j \neq i} b_j}$, as functions of s_i and $s_{\max}$.

Claim 1. When $s_{\max} < \frac{1}{2}$, we have:

$$s_i \leq \frac{b_i}{\sum_{j \neq i} b_j} \leq \left(1 + \frac{s_{\max}}{1 - 2s_{\max}}\right) s_i \quad (25)$$

Claim 1 implies that the coefficient on c_i in i 's bid in equilibrium is within a factor

$$\left(1 + \frac{s_{\max}}{1 - 2s_{\max}}\right)$$

of her capacity share, s_i . Thus, when $s_{\max}$ is small, c_i coefficients are approximately equal to capacity shares. Intuitively, when markets are relatively competitive, an agent who constitutes roughly 10% of the market's storage capacity will increase her spot market bids by approximately 10% of the size of her contract position. The factor s_i thus generalizes the factor $\frac{1}{n-1}$ in (7), the expression for spot market bid curves in the baseline model.

If all agents' contract positions have equal variance, σ_c^2 , we can also derive bounds for manipulation-induced basis risk. Define the capacity HHI (Herfindahl-Hirschman index) as:

$$HHI \equiv \sum_{i=1}^n s_i^2 \quad (26)$$

The HHI is a standard measure of market concentration used in antitrust analysis. When agents are symmetric, $\kappa_i = \kappa$, the HHI is simply $\frac{1}{n}$.

(b) Capacity shares, HHI, and basis-risk bounds

Original paper excerpt:

Interpretation: The coefficient on c_i is larger for traders with more spot-market storage capacity. Traders who can absorb more spot goods have more ability to move benchmark prices, so their derivative positions are more relevant for manipulation risk.

5.2 Capacity shares and concentration

Let trader i 's capacity share be

$$s_i = \frac{\kappa_i}{\sum_{j=1}^n \kappa_j}.$$

The paper relates manipulation risk to capacity concentration through an HHI-like object,

$$HHI = \sum_{i=1}^n s_i^2.$$

When the largest capacity share is not too large, the contribution of contract positions to basis risk can be bounded using capacity shares and HHI .

Interpretation: In asymmetric markets, not every spot trader is equally important. A position held by a large-capacity spot trader matters more than the same position held by a small-capacity trader.

5.3 Inventory shocks

The second extension adds inventory shocks y_i . The spot payoff becomes

$$W_{spot,i}(c_i, z_i, y_i, p) = pc_i + \psi z_i - \frac{(z_i + y_i)^2}{2\kappa_i} - pz_i.$$

Here y_i shifts the trader's marginal value of the spot good. Inventory shocks represent genuine spot-market needs or preference shocks, not manipulation.

for the spot good.

5.2. Inventory shocks

Next, we add inventory shocks to the model. Assume spot traders' wealth is:

$$W_{spot,i}(c_i, z_i, y_i, p) = \underbrace{pc_i}_{\text{Contract payoff}} + \underbrace{\psi z_i - \frac{(z_i + y_i)^2}{2\kappa_i}}_{\text{Spot good payoff}} - \underbrace{pz_i}_{\text{Spot good price}} \quad (29)$$

Expression (29) assumes that agents enter the spot market with some existing inventory position, y_i , in the spot good.¹⁶ I assume that each y_i has mean 0 and full support, but otherwise can be arbitrarily distributed; inventory shocks can be asymmetrically distributed across agents, and can be arbitrarily correlated with other agents' inventory shocks and contract positions. Inventory shocks imply that agents will trade the spot good even if they do not hold contract positions, so that not all trade volume in the spot market is caused by manipulation.

Proposition 3. When spot traders' wealth is described by (29), there is a unique linear ex-post equilibrium in the general

¹⁶ Inventory shocks y_i can alternatively be interpreted as preference shocks for the spot good: expanding the quadratic term in (29), we have:

model, in which i submits the bid curve:

$$z_{Bi}(p; y_i, c_i) = -\frac{b_i}{\kappa_i} y_i + \frac{b_i}{\sum_{j \neq i} b_j} c_i - b_i(p - \psi) \quad (30)$$

The spot auction price is:

$$p - \psi = \frac{1}{\sum_{i=1}^n b_i} \sum_{i=1}^n \left[-\frac{b_i}{\kappa_i} y_i + \frac{b_i}{\sum_{j \neq i} b_j} c_i \right] \quad (31)$$

bid slopes b_i satisfy:

$$b_i = \frac{B + 2\kappa_i - \sqrt{B^2 + 4\kappa_i^2}}{2} \quad (32)$$

and $B \equiv \sum_{i=1}^n b_i$ is the unique positive solution to the equation:

$$B = \sum_{i=1}^n \frac{2\kappa_i + B - \sqrt{B^2 + 4\kappa_i^2}}{2} \quad (33)$$

First, I specialize [proposition 3](#) to the case in which agents are fully symmetric.

Corollary 1. Suppose that $\kappa_i = \kappa$, $\text{Var}(y_i) = \sigma_y^2$, $\text{Var}(c_i) = \sigma_c^2$ for all i , and all y_i and c_i are independent. Spot traders' equilibrium bids are:

$$z_{Bi}(p; y_i, c_i) = -\frac{n-2}{n-1} y_i + \frac{1}{n-1} c_i - \frac{n-2}{n-1} \kappa(p - \psi) \quad (34)$$

Price variance is:

(a) Proposition 3: inventory shocks in the general model

(b) Symmetric case with inventory and contract shocks

Original paper excerpt:

Read:

- Inventory shocks enter bid curves because they change spot-market fundamentals.
- Contract positions enter bid curves because they change derivative payoffs.
- In symmetric markets, inventory shocks pass through to bids with coefficient roughly $-(n-2)/(n-1)$.
- Contract positions pass through with coefficient $1/(n-1)$.
- As the spot market becomes more competitive, contract-position-driven bidding becomes less important.

Economic message: The model distinguishes legitimate spot trading from manipulation. Inventory shocks are a reason to trade the spot good. Contract positions are a reason to move the benchmark.

5.4 Perfect hedging of inventory and the non-manipulative benchmark

A special case is $c_i = -y_i$: the trader's contract position exactly offsets the inventory position. Then the contract-driven and inventory-driven components of bids cancel in a way that makes the trade non-manipulative.

Interpretation: This gives a practical conceptual benchmark. A spot trader whose derivative position exactly hedges a spot exposure may still trade spot goods around settlement, but those trades need not be manipulative if they simply pass through genuine inventory needs.

5.5 Measuring manipulation-induced basis risk

The paper gives an empirical measurement formula. Define coefficient vectors $\mathbf{k}_c$ and $\mathbf{k}_y$ from equilibrium bid slopes and storage capacities, and define covariance matrices for contract positions and inventory shocks:

$$\Sigma_{cc}, \quad \Sigma_{yy}, \quad \Sigma_{yc}.$$

Proposition 4 states that in the general model,

$$\text{Var}(p - \psi) = \frac{1}{B^2} (\mathbf{k}_y' \Sigma_{yy} \mathbf{k}_y - 2\mathbf{k}_y' \Sigma_{yc} \mathbf{k}_c + \mathbf{k}_c' \Sigma_{cc} \mathbf{k}_c).$$

Interpretation: Regulators do not need to read traders' minds. They can estimate manipulation risk from observed bid slopes and the covariance structure of contract positions and spot-market inventory shocks.

5.6 Physical delivery contracts

The baseline model uses cash settlement. The paper then shows that physical delivery contracts behave identically within the model when traders can submit market-on-expiration orders.

Read:

- With competitive bidding and physical delivery, spot traders close out their contracts before settlement and no physical delivery occurs.
- The spot price equals ψ , so the hedger perfectly hedges.
- In strategic equilibrium, the net delivery quantities and settlement prices are identical to the baseline cash-settlement case.
- A long physical-delivery contract holder manipulates by closing less of her position than she would competitively, thereby taking delivery of some spot goods and affecting the settlement price.

Economic message: The manipulation problem is not unique to cash settlement. It is a more general consequence of linking derivative payoffs to a settlement price created by strategic spot-market trades.

6 Regulatory implications

6.1 A behavioral definition of manipulation

The paper proposes a definition based on behavior rather than legal intent. Manipulation is the part of a spot-market bid that is caused by the trader's derivative-contract position. This component can be distinguished from two legal strategic behaviors:

- lowering liquidity provision because of price impact;
- shading bids because inventory shocks affect residual supply.

could be empirically estimated by a regulator. To ensure that basis risk is finite, we assume that all y_i and c_i have finite variances and covariances. Expression (31) shows that auction prices are linear in agents' inventory and contract positions: this implies that the moments of auction prices are functions of the moments of y_i and c_i . As in proposition 3, we define $B \equiv \sum_{i=1}^n b_i$ as the sum of agents' bid slopes b_i . Define the coefficient vectors $\mathbf{k}_c, \mathbf{k}_y$ as:

$$\mathbf{k}_c = \begin{pmatrix} \frac{b_1}{\sum_{j \neq 1} b_j} \\ \vdots \\ \frac{b_n}{\sum_{j \neq n} b_j} \end{pmatrix}, \mathbf{k}_y = \begin{pmatrix} \frac{b_1}{\kappa_1} \\ \vdots \\ \frac{b_n}{\kappa_n} \end{pmatrix}$$

Define the covariance matrices of agents' contract position, Σ_{cc} , agents' inventory positions, Σ_{yy} , and the inventory-contract covariance matrix Σ_{yc} , respectively, as matrices with elements:

$$\begin{aligned} \Sigma_{cc}(i, j) &= \text{Cov}(c_i, c_j), \quad \Sigma_{yy}(i, j) \\ &= \text{Cov}(y_i, y_j), \quad \Sigma_{yc}(i, j) = \text{Cov}(y_i, c_j) \end{aligned}$$

Proposition 4. In the general model of proposition 3, basis risk is:

$$\text{Var}(p - \psi) = \frac{1}{\kappa} (\mathbf{k}_v' \Sigma_{vv} \mathbf{k}_v - 2 \mathbf{k}_v' \Sigma_{vc} \mathbf{k}_c + \mathbf{k}_c' \Sigma_{cc} \mathbf{k}_c) \quad (39)$$

Expanding,

and ignoring terms that do not depend on c_i or z_i , we have:

$$c_i \psi + z_i \psi - \frac{1}{2\kappa} (z_i^2 + 2z_i c_i + c_i^2) - p z_i \quad (43)$$

Expression (40) shows that the hedger and the market maker make and take no deliveries of the spot good. Thus, in order for spot markets to clear, the total quantities that spot traders take deliveries of – which are equal to the quantities z_i purchased in the spot market, plus traders' total contract positions c_i – must sum to 0:

$$\sum_{i=1}^n z_i(p; c_i, \psi) + c_i = 0$$

As above, we can define the quantity spot trader i takes delivery of, q_i , as the sum of z_i and c_i :

$$q_i(p; c_i, \psi) \equiv z_i(p; c_i, \psi) + c_i \quad (44)$$

Thus, when $z_i = -c_i$, trader i does not take any deliveries: she closes all her contract positions just before settlement. When $z_i = 0$, trader i takes delivery of her entire contract position c_i . When z_i is between 0 and $-c_i$, i closes some fraction of her contract position, and takes delivery of the remainder.

Proposition 5 describes outcomes in the spot market in a competitive-bidding benchmark, and proposition 6 describes behavior in strategic equilibrium.

Proposition 5. With physical delivery contracts, if spot traders behave competitively, as if residual supply in the spot market is perfectly elastic, spot traders' bids are:

$$z_{Bi}(p; c_i, \psi) = -c_i + \kappa(p - \psi) \quad (45)$$

Spot traders' aggregate demand for deliveries of the spot good is:

$$\sum_{i=1}^n q_i(p; c_i, \psi) = \sum_{i=1}^n z_{Bi}(p; c_i, \psi) + c_i = \sum_{i=1}^n \kappa(p - \psi) \quad (46)$$

Hence, the spot market clears with no trade: spot traders do not make or take delivery of the spot good, and the market clearing price is $p = \psi$.

Proposition 6. In strategic equilibrium with physical delivery contracts, spot traders' optimal bids in the spot market are:

$$z_{Bi}(p; c_i, \psi) = -\frac{n-2}{n-1} c_i + \frac{n-2}{n-1} \kappa(p - \psi) \quad (47)$$

The quantity of the spot good that traders take delivery of, q_i , is identical to spot bids in the baseline model, (7) of proposition 1:

$$q_i(p; c_i, \psi) = \frac{1}{n-1} c_i - \frac{n-2}{n-1} \kappa(p - \psi) \quad (48)$$

The market clearing price is identical to (8) of proposition 1 in the baseline model.

tools, such as position limits, can be used to reduce market participants' incentives to manipulate.

Defining manipulation, and behavioral solutions. As subsection 2.2 discusses, contract market manipulation is illegal, but the law does not precisely define manipulation. Courts, regulators and academics have faced substantial difficulty in precisely defining manipulation, and this is a major barrier to enforcing manipulation law. This paper presents a potential definition of contract market manipulation. When strategic spot market traders hold derivative contracts linked to spot markets, traders have incentives to modify their trading behavior, to increase the payoffs on their contract positions. Within the model, this force can be distinguished from two other incentives traders in imperfect markets have to distort trading behavior: decreasing liquidity provision, that is, lowering the slopes of their bid curves relative to a competitive benchmark; and "shading bids", that is, lowering the extent to which inventory shocks affect bids.

If spot traders were barred from holding derivative contracts, they would decrease liquidity provision and shade bids, but would not have incentives to "manipulate" in the sense of trading spot goods to increase contract payoffs. In most markets, it is likely impractical to bar spot traders from holding derivative positions. However, the thought experiment of eliminating spot traders' derivative positions may be useful as a conceptual benchmark for arguing that illegal contract market manipulation is occurring: regulators might ask whether a given pattern of spot good trades could have been profitable, if a trader did not hold derivative contracts.

This definition also shows what trading patterns regulators might look for, to demonstrate that market participants are engaging in illegal manipulation. Trading in spot markets, which could not be profitable on a standalone basis, may be profitable once payoffs from contract positions are taken into account. The model also shows that traders may "over-hedge" in equilibrium, buying contract positions much larger than their factor exposures, deliberately building up exposure to factor risk, since they know they can manipulate contract settlement prices.

Some of the features in the model may be difficult for regulators to bring to data. For example, while inventory or preference shocks are precisely defined within the model, these may be difficult to estimate in practice. Market participants who are accused of manipulation could thus always claim that their spot trades are driven by inventory or preference shocks, rather than the desire to manipulate. The paper's results suggest that over-hedging is symptomatic of manipulation, but demonstrating that agents are over-hedging requires estimating the size of market participants' factor risk exposures, which may be difficult if market participants have positions in many different products with correlated values. Moreover, market participants may enter into positions to speculate rather than hedge risk, and a market participant could always argue that a posi-

efforts by shedding light on the forces that affect manipulation risk. Manipulation risk depends on the size of spot traders' contract positions. Regulators could impose position limits on spot traders' contract positions, especially close to contract settlement. Position limits are currently imposed in many derivative contract markets, but their primary purpose seems to be preventing excessive price volatility.²¹ As a result, the CFTC currently applies position limits more harshly for pure financial market participants than spot traders.²² The results of this paper suggest that, for the purposes of alleviating manipulation risk, position limits should actually be more harsh for spot traders, since they have a greater ability to trade spot goods to influence settlement prices.

Spot market storage capacity and competitiveness also affect manipulation risk.²³ This suggests, for example, that a recent regulatory effort to move interest rate benchmarks to larger and more competitive underlying markets could potentially alleviate manipulation risk.²⁴ Moreover, this implies that regulators cannot determine manipulation incentives by focusing on derivative contract markets in isolation. Contract market regulators should monitor storage capacity and concentration in spot markets, perhaps in collaboration with antitrust and spot market regulators, as spot market structure is an important factor affecting contract market manipulation risk.

Subsection 5.3 of the paper shows how to estimate the size of manipulation-induced market distortions, using data which regulators observe in many markets. Regulators could use these measures to decide whether to approve new contracts, and how large contract position limits should be. These measures would also inform regulators how changes in spot market structure, such as mergers or changes in aggregate storage capacity, would influence ma-

(a) Defining manipulation relative to other strategic behavior

(b) Structural regulation and measurement implications

Original paper excerpt:

Interpretation: If spot traders were barred from holding derivative contracts, they would still shade bids and provide less liquidity, but they would no longer have incentives to move the settlement price to improve derivative payoffs. That thought experiment identifies the manipulation component.

6.2 Position limits

Because basis risk rises with spot traders' derivative positions, position limits can reduce manipulation risk. The paper emphasizes that the relevant limits should apply more strongly to traders who can move spot prices: spot-market participants with storage capacity, large balance sheets, or concentrated spot-market power.

Interpretation: Traditional position limits often target excessive speculation or volatility. In this model, the key criterion is not speculation per se. It is the ability of a trader's contract position to influence spot-market bidding at settlement.

6.3 Spot-market competition policy

Increasing competition among spot-market participants reduces manipulation risk. Holding aggregate capacity fixed, splitting capacity across more traders lowers each trader's price impact and reduces the effect of derivative positions on settlement bids.

Interpretation: Contract-market regulation should not look only at the contract market. The underlying spot market's concentration and storage capacity determine whether the benchmark can be manipulated.

6.4 Benchmark design

The model studies a uniform-price double auction, but the implications extend to many benchmark-setting mechanisms. If a benchmark is based on prices in a market where traders have price impact, derivative positions can create manipulation incentives.

Interpretation: Benchmark reform can lower manipulation risk by using larger, deeper, more competitive, or more robust benchmark mechanisms. But benchmark design and contract position limits are complements, not substitutes.

7 Tables, figures, and original-paper excerpts

The paper contains one plotted figure and no empirical tables. The key visual material is therefore the original Figure 1 plus cropped proposition/equation excerpts that define the model and results.

7.1 Model structure and baseline equilibrium

Read:

- The five-stage game clarifies when information and incentives arrive.
- Factor exposures are private at the time of derivative trade.
- The risk factor is common knowledge before spot bidding.
- Settlement occurs after the spot auction, so spot bids directly affect contract payoffs.
- Proposition 1 is the central equilibrium characterization.

Economic message: The model turns manipulation into an equilibrium response to contract-settlement design.

Agents play a 5-stage game.

1. The hedger and spot traders draw their factor risk exposures, respectively, x_H and $x_1 \dots x_n$.
2. The hedger and spot traders submit market orders, respectively $c_H(x_H)$ and $c_i(x_i)$, to trade derivative contracts with the market maker.
3. The risk factor, ψ , is realized.
4. Spot traders bid to trade the spot good in an auction.
5. Derivative contracts pay all contract holders based on the spot auction price.

In stage 1, the hedger is endowed with x_H units of a productive asset, and each spot trader i is endowed with x_i units of the productive asset, where:

$$x_H \sim N(0, \sigma_H^2), \quad x_i \sim N(0, \sigma_x^2)$$

A position x_i in the productive asset pays its holder $x_i \psi$ net units of wealth. Thus, the value of each unit of the productive asset depends on a normally distributed risk factor ψ .

(a) Model timing and risk exposure setup

sitions; thus, colluding spot traders, who coordinate their spot market bids to maximize joint profits, would manipulate more per unit contract than independently optimizing traders. This is an interesting direction to explore, but I leave this to future work.

4. Equilibrium

Proposition 1 describes equilibrium values of spot traders' bid curves, auction prices, contract purchases, and expected utility for all agents.

Proposition 1. For any $\alpha, \mu_\psi, \sigma_\psi^2, \kappa, \sigma_x^2, \sigma_H^2, n$, there is a unique equilibrium, in which spot traders' spot market bids and contract purchasing strategies are linear and symmetric across agents. In stage 1, the market maker sets contract price $p_c = \mu_\psi$. Spot traders' bid curves in the spot market are:

$$z_{Bi}(p; c_i, \psi) = \frac{1}{n-1} c_i - \frac{n-2}{n-1} \kappa (p - \psi) \quad (7)$$

Spot auction prices are:

$$p - \psi = \frac{\sum_{i=1}^n c_i}{n(n-2)\kappa} \sim N\left(0, \frac{\sigma_c^2}{n(n-2)^2 \kappa^2}\right) \quad (8)$$

Spot traders' contract positions $c_i(x_i)$ are linear in traders' factor exposures x_i , so spot traders' contract positions are normally distributed with mean 0 and variance σ_c^2 , where:

$$c_i(x_i) = -tx_i, \quad \sigma_c^2 = t^2 \sigma_x^2 \quad (9)$$

where t satisfies:

$$t \equiv \left(1 + \frac{\alpha \sigma_\eta^2 - \kappa}{(\alpha \sigma_\psi^2 \kappa)((n^2 - 2n)\kappa + \alpha \sigma_\eta^2)}\right)^{-1} \quad (10)$$

and σ_η^2 is the unique positive value satisfying:

(b) Baseline equilibrium in Proposition 1

7.2 Basis risk and welfare

Read:

- Basis risk is the settlement price error $p - \psi$.
- The formula shows how basis risk depends on contract-position variance, competition, and capacity.
- Figure 1 shows that equilibrium spot-trader welfare can be below competitive welfare.
- The welfare-maximizing hedging aggressiveness can be lower than the equilibrium hedging aggressiveness.

Economic message: Manipulation creates a strategic externality: each trader wants to move the price, but all traders suffer from the resulting benchmark noise.

Since auction prices depend on spot traders' contract positions c_i , which depend on traders' random factor exposures x_i , the equilibrium auction price p is a noisy signal of the risk factor ψ . The difference, $p - \psi$, is normally distributed, and expression (8) characterizes its variance:

$$\text{Var}(p - \psi) = \frac{\sigma_c^2}{n(n-2)^2\kappa^2} \quad (16)$$

I call this variance *manipulation-induced basis risk*.

Comparative statics of basis risk. We will define three quantities that influence the level of basis risk. First, define the aggregate *storage capacity* of the market as $n\kappa$, the sum over all n spot traders' storage capacities. $n\kappa$ would be the slope of aggregate market demand for the spot good, if all agents submitted bid curves as if they had no price impact, as in (6). Second, define the level of market *competition* simply as n , the number of spot traders present in the market. Third, define the size of contract positions as σ_c^2 , the variance of an individual spot trader's contract position. This is an equilibrium object in the model, but in practice it can often be observed directly by regulators, and directly influenced using policies such as contract position limits.

Basis risk decreases to 0 as storage capacity becomes infinitely large, $n\kappa \rightarrow \infty$, holding fixed n and σ_c^2 . Intuitively, when $n\kappa$ is large, the price impact of manipulative spot trading decreases towards 0, causing prices converge to ψ . However, while *prices* become competitive, spot market *quantities* do not become competitive. From (7), the coefficient on contract positions in agents' bids is $\frac{1}{n-1}$, independent of κ . If κ increases holding n fixed, prices converge to ψ , but spot market bids, and thus allocations in the spot market, are still distorted in the limit.

Basis risk also decreases towards 0 as markets become competitive, $n \rightarrow \infty$, holding fixed $n\kappa$ and σ_c^2 . This is be-

(a) Basis risk and comparative statics

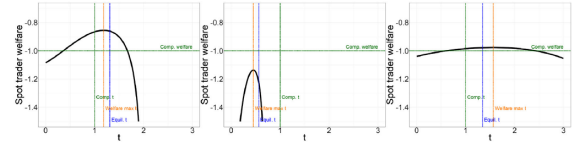


Fig. 1. Spot trader welfare. Notes. Expected welfare of spot traders, (13), as a function of the hedging aggressiveness parameter t , for three different sets of parameters. The green vertical and horizontal lines denote the competitive values of t and welfare, which are always equal to 1 and -1 respectively. The blue line denotes the unique equilibrium value of t , and the orange line denotes the value of t which maximizes spot traders' welfare. The parameters for the left panel are $n=3, \alpha=1, \kappa=0.5, \sigma_c^2=0.3, \sigma_x^2=0.5$; for the middle panel, they are $n=5, \alpha=1, \kappa=0.05, \sigma_c^2=0.9, \sigma_x^2=1$; for the right panel, they are $n=3, \alpha=1, \kappa=0.8, \sigma_c^2=1.5, \sigma_x^2=0.05$.

(b) Spot trader welfare under different parameters

7.3 Extensions and measurement

Read:

- Asymmetric capacity changes which traders matter most for manipulation risk.
- Inventory shocks identify legitimate spot-market trading needs.
- Proposition 4 gives a regulator-facing formula for basis risk.
- Physical delivery and cash settlement are equivalent in the model under market-on-expiration orders.

Economic message: The extensions make the model operational. They show how to map observed market data into a measure of benchmark vulnerability.

pendix B.4 discusses, spot traders may buy contracts to speculate rather than hedge, or as online appendix B.5 discusses, spot traders' contract purchases may have price impact. A spot trader who holds a long contract position profits from higher spot prices, regardless of why she entered into the contract in the first place. Thus, spot traders' contract positions are an observable sufficient statistic for spot traders' incentives to manipulate spot markets.

The following proposition characterizes spot good bids and prices in the unique equilibrium of the general model.

Proposition 2. When spot traders' wealth is described by (19), there is a unique linear ex-post equilibrium, in which i submits the bid curve:

$$z_{Bi}(p; c_i) = \frac{b_i}{\sum_{j \neq i} b_j} c_i - b_i(p - \psi) \quad (20)$$

The spot auction price is:

$$p - \psi = \frac{1}{\sum_{i=1}^n b_i} \sum_{i=1}^n \left[\frac{b_i}{\sum_{j \neq i} b_j} c_i \right] \quad (21)$$

bid slopes b_i satisfy:

$$b_i = \frac{B + 2\kappa_i - \sqrt{B^2 + 4\kappa_i^2}}{2} \quad (22)$$

and $B \equiv \sum_{i=1}^n b_i$ is the unique positive solution to the equation:

$$B = \sum_{i=1}^n \frac{2\kappa_i + B - \sqrt{B^2 + 4\kappa_i^2}}{2} \quad (23)$$

Manipulation incentives depend most directly on the coefficient on c_i , $\frac{b_i}{\sum_{j \neq i} b_j}$, in expression (20). From (22), b_i is increasing in κ_i . So agents with larger storage capacities

(a) Asymmetric agents

for the spot good.

5.2. Inventory shocks

Next, we add inventory shocks to the model. Assume spot traders' wealth is:

$$W_{spot,i}(c_i, z_i, y_i, p) = \underbrace{pc_i}_{\text{Contract payoff}} + \underbrace{\psi z_i - \frac{(z_i + y_i)^2}{2\kappa_i}}_{\text{Spot good payoff}} - \underbrace{pz_i}_{\text{Spot good price}} \quad (29)$$

Expression (29) assumes that agents enter the spot market with some existing inventory position, y_i , in the spot good.¹⁶ I assume that each y_i has mean 0 and full support, but otherwise can be arbitrarily distributed; inventory shocks can be asymmetrically distributed across agents, and can be arbitrarily correlated with other agents' inventory shocks and contract positions. Inventory shocks imply that agents will trade the spot good even if they do not hold contract positions, so that not all trade volume in the spot market is caused by manipulation.

Proposition 3. When spot traders' wealth is described by (29), there is a unique linear ex-post equilibrium in the general

¹⁶ Inventory shocks y_i can alternatively be interpreted as preference shocks for the spot good: expanding the quadratic term in (29), we have:

(b) Inventory shocks

could be empirically estimated by a regulator. To ensure that basis risk is finite, we assume that all y_i and c_i have finite variances and covariances. Expression (31) shows that auction prices are linear in agents' inventory and contract positions: this implies that the moments of auction prices are functions of the moments of y_i and c_i . As in proposition 3, we define $B \equiv \sum_{i=1}^n b_i$ as the sum of agents' bid slopes b_i . Define the coefficient vectors $\mathbf{k}_c, \mathbf{k}_y$ as:

$$\mathbf{k}_c = \begin{pmatrix} \frac{b_1}{\sum_{j \neq 1} b_j} \\ \vdots \\ \frac{b_n}{\sum_{j \neq n} b_j} \end{pmatrix}, \mathbf{k}_y = \begin{pmatrix} \frac{b_1}{\kappa_1} \\ \vdots \\ \frac{b_n}{\kappa_n} \end{pmatrix}$$

Define the covariance matrices of agents' contract position, Σ_{cc} , agents' inventory positions, Σ_{yy} , and the inventory-contract covariance matrix Σ_{yc} , respectively, as matrices with elements:

$$\begin{aligned} \Sigma_{cc}(i, j) &= \text{Cov}(c_i, c_j), \quad \Sigma_{yy}(i, j) \\ &= \text{Cov}(y_i, y_j), \quad \Sigma_{yc}(i, j) = \text{Cov}(y_i, c_j) \end{aligned}$$

Proposition 4. In the general model of proposition 3, basis risk is:

$$\text{Var}(p - \psi) = \frac{1}{\kappa} (\mathbf{k}'_y \Sigma_{yy} \mathbf{k}_y - 2 \mathbf{k}'_y \Sigma_{yc} \mathbf{k}_c + \mathbf{k}'_c \Sigma_{cc} \mathbf{k}_c) \quad (39)$$

(a) Measurement of basis risk

Expanding, and ignoring terms that do not depend on c_i or z_i , we have:

$$c_i \psi + z_i \psi - \frac{1}{2\kappa} (z_i^2 + 2z_i c_i + c_i^2) - p z_i \quad (43)$$

Expression (40) shows that the hedger and the market maker make and take no deliveries of the spot good. Thus, in order for spot markets to clear, the total quantities that spot traders take deliveries of – which are equal to the quantities z_i purchased in the spot market, plus traders' total contract positions c_i – must sum to 0:

$$\sum_{i=1}^n z_i(p; c_i, \psi) + c_i = 0$$

As above, we can define the quantity spot trader i takes delivery of, q_i , as the sum of z_i and c_i :

$$q_i(p; c_i, \psi) \equiv z_i(p; c_i, \psi) + c_i \quad (44)$$

Thus, when $z_i = -c_i$, trader i does not take any deliveries: she closes all her contract positions just before settlement. When $z_i = 0$, trader i takes delivery of her entire contract position c_i . When z_i is between 0 and $-c_i$, i closes some fraction of her contract position, and takes delivery of the remainder.

Proposition 5 describes outcomes in the spot market in a competitive-bidding benchmark, and proposition 6 describes behavior in strategic equilibrium.

Proposition 5. With physical delivery contracts, if spot traders behave competitively, as if residual supply in the spot market is perfectly elastic, spot traders' bids are:

$$z_{Bi}(p; c_i, \psi) = -c_i + \kappa(p - \psi) \quad (45)$$

Spot traders' aggregate demand for deliveries of the spot good is:

$$\sum_{i=1}^n q_i(p; c_i, \psi) = \sum_{i=1}^n z_{Bi}(p; c_i, \psi) + c_i = \sum_{i=1}^n \kappa(p - \psi) \quad (46)$$

Hence, the spot market clears with no trade: spot traders do not make or take delivery of the spot good, and the market clearing price is $p = \psi$.

Proposition 6. In strategic equilibrium with physical delivery contracts, spot traders' optimal bids in the spot market are:

$$z_{Bi}(p; c_i, \psi) = -\frac{n-2}{n-1} c_i + \frac{n-2}{n-1} \kappa(p - \psi) \quad (47)$$

The quantity of the spot good that traders take delivery of, q_i , is identical to spot bids in the baseline model, (7) of proposition 1:

$$q_{Bi}(p; c_i, \psi) = \frac{1}{n-1} c_i - \frac{n-2}{n-1} \kappa(p - \psi) \quad (48)$$

The market clearing price is identical to (8) of proposition 1 in the baseline model.

(b) Physical-delivery contracts

8 Short summary

- The paper studies manipulation in derivative contract markets whose payoff is linked to spot-market settlement prices.

- Agents want to hedge a common risk factor ψ , but contracts settle on a spot-market price p .
- If the spot market were competitive, $p = \psi$ and derivatives would perfectly share risk.
- Spot traders have price impact, so derivative positions affect their spot-market bids.
- Long contract positions induce spot purchases; short contract positions induce spot sales.
- These trades add non-fundamental noise to the settlement price.
- The variance $\text{Var}(p - \psi)$ is manipulation-induced basis risk.
- Basis risk falls with greater storage capacity, more spot-market competition, and smaller derivative positions held by spot traders.
- Manipulation can reduce the welfare of both hedgers and spot traders.
- The model gives a precise definition of manipulation: spot trades motivated by derivative payoffs.
- Extensions allow asymmetric spot traders, inventory shocks, empirical measurement, and physical delivery.
- Regulators can use bid slopes, contract positions, and inventory covariances to estimate manipulation risk.
- Position limits and spot-market competition policy can reduce manipulation risk.

9 Conclusion

9.1 Core conclusion

Zhang shows that manipulation in derivative contract markets is not merely a legal-intent problem. It is an economic market failure that arises when derivative payoffs are tied to spot prices produced by strategic traders. The same trader who holds a derivative position can trade in the spot market to move the settlement price, creating basis risk for all contract users.

9.2 Economic intuition

Derivatives are useful because they allow cheap hedging. Spot markets are useful because their prices reveal fundamentals. The problem is that the settlement benchmark turns a spot price into a payoff-relevant object for derivative traders. Once spot traders know their derivative positions, they trade in the spot market partly to change contract payoffs rather than only to satisfy genuine spot-market needs. This makes the benchmark noisier and undermines the hedging function of the contract.

9.3 Takeaway

The paper gives regulators two practical insights. First, manipulation should be defined as the derivative-position-driven component of spot-market trading, not as all intentional price impact. Second, manipulation risk can be measured and regulated using observable market structure: storage capacity, spot-market concentration, derivative-position size, bid slopes, and covariance between spot needs and contract positions.